

1. Given two strings a and b, return true if a can be constructed by using the letters from b and false otherwise. Each letter in b can only be used once in a.

Example 1:

Input: a = "a", b = "b"

Output: false

Example 2:

Input: a = "aa", b = "ab"

Output: false

Example 3:

Input: a = "aa", b = "aab"

Output: true

2. Given two non-negative integers, num1 and num2 represented as string, return the sum of num1 and num2 as a string. You must solve the problem without using any built-in library for handling large integers (BigInteger). You must also not convert the inputs to integers directly.

Example 1:

Input: num1 = "11", num2 = "123"

Output: "134"

Example 2:

Input: num1 = "456", num2 = "77"

Output: "533"

Example 3:

Input: num1 = "0", num2 = "0"

Output: "0"

3. A peak element is an element that is strictly greater than its neighbors. Given a 0-indexed integer array nums, find a peak element, and return its index. If the array contains multiple peaks, return the index to any of the peaks.

Example 1:

Input: nums = [1,2,3,1]

Output: 2

Explanation: 3 is a peak element and you should return the index number 2.

Example 2:

Input: nums = [1,2,1,3,5,6,4]

Output: 5

Explanation: You can return either index number 1 where the peak element is 2, or index number 5 where the peak element is 6.

4. Given an array nums of size n, return the majority element. The majority element is the element that appears more than $n / 2$ times.

Example 1:

Input: nums = [3,2,3]

Output: 3

Example 2:

Input: nums = [2,2,1,1,1,2,2]

Output: 2

5. Given an integer array `nums`, return an array `answer` such that `answer[i]` is equal to the product of all the elements of `nums` except `nums[i]`.

Example 1:

Input: `nums = [1,2,3,4]`

Output: `[24,12,8,6]`

Example 2:

Input: `nums = [-1,1,0,-3,3]`

Output: `[0,0,9,0,0]`